Discovering the mysterious fungi of Esker Forests: a mycological madness?

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Introduction

- Eskers are **fluvioglacial geological formations** formed by the passage of glaciers.
- High **ecological** and **economic** value (e.g drinking water quality, *Pinus banksiana, Tricholoma magnivelare,* gravel and sand).
- **No research focusing on Fungi of Northern Quebec Esker Forests has been carried out to date...**

- The study of mycodiversity is **essential** in our understanding of **ecosystem functioning**
- Esker Forests represent a **particular environment** for the biodiversity they contain.
- A **unique**, **vulnerable** ecosystem, and associated fungi under **high anthropogenic pressures**, including logging, mining and water exploitation.
- Impacted **ethnomycological activities** of local autochthone communities.



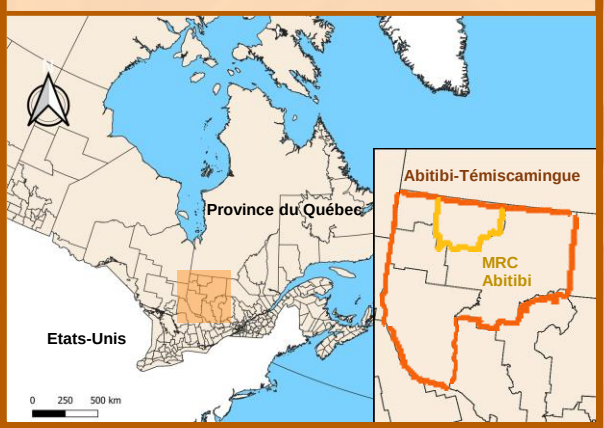


Objectives

- **Describe** soil fungal communities assembly of different Esker Forests.
- **Identify** carpophore-producing fungal species present in Esker Forests.

Methods

Study area



DNA Metabarcoding (eDNA)

I. **DNA extraction**, **PCR**, and soil samples **sequencing**

II. **Comparative analysis** of fungal communities + Calculation of biodiversity indices and statistical analysis

R studio



Field inventories





I. Macroscopic and microscopic **identification** and **description**

II. **DNA extraction, PCR,** and collected carpophore **sequencing** (DNA Barcoding)



Contributions

- **First study** to assess the fungal diversity of Esker Forests.
- Establish **robust taxonomic knowledge** on Esker Forests fungi.
- **Promote** these ecosystems and the related Fungi.

Reference: Kim, S., Axelsson, E. P., Girona, M. M., & Senior, J. K. (2021). Continuous-cover forestry maintains soil fungal communities in Norway spruce dominated boreal forests. Forest Ecology and Management, 480, 118659. https://doi.org/10.1016/j.foreco.2020.118659

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